**Unit:** Unit 0: Foundations of Algebra
**Topic:** Isolating Variables

**Name:** ________________
**Date:** ___________

# Homework (Habanero)

**Q1.** Solve for $x$: $2x + 5 = 11$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

**Q2.** Rearrange to isolate $y$: $y - 3 = 2x + 1$
(A) $y = 2x + 4$
(B) $y = 2x - 4$
(C) $y = x + 4$
(D) $y = x - 4$
(E) $y = 3x + 4$

**Q3.** Solve for $z$: $z/4 = 9 - 3x$
(A) $z = 36 - 12x$
(B) $z = 36 + 12x$
(C) $z = 9 - 3x$
(D) $z = 4(9 - 3x)$
(E) $z = 12x - 36$

**Q4.** A patient's dosage of medicine is given by $D = 2t+5$, where $t$ is time in hours. Solve for $t$
(A) $t = (D - 5)/2$
(B) $t = (D + 5)/2$
(C) $t = 2D - 5$
(D) $t = 2D + 5$
(E) $t = D/2 - 5$

**Q5.** Isolate $x$ in $x/2 + 2 = 7$
(A) $x = 10$
(B) $x = 12$
(C) $x = 14$
(D) $x = 16$
(E) $x = 18$

**Q6.** Rearrange to solve for $r$: $A = pir^2$
(A) $r = sqrtA/pi$
(B) $r = sqrtA/pi$
(C) $r = A/pi$
(D) $r = Api$
(E) $r = pi/A$

**Q7.** Solve for $p$ in $2p + 8 = 3p - 4$
(A) $p = 12$
(B) $p = 10$
(C) $p = 8$
(D) $p = 6$
(E) $p = 4$

**Q8.** The growth rate of a cell is given by $N = 2t + N_0$, where $N_0$ is the initial number of cells. Solve for $t$
(A) $t = (N - N_0)/2$
(B) $t = (N + N_0)/2$
(C) $t = N/2 - N_0$
(D) $t = 2N - N_0$
(E) $t = N - 2N_0$

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**Open Ended Questions (FRQ)**

**Q9.** A new medicine is being tested, and its effectiveness is modeled by $E = 3d+2$, where $d$ is the dosage. Solve for $d$ and explain your steps

**Q10.** The equation $pV = nRT$ relates the pressure $p$ of a gas to its volume $V$, number of moles $n$, gas constant $R$, and temperature $T$. Rearrange to solve for $V$ and give an example

**Chef's Tips**



- Read each question carefully
- Use the correct formula and units
- Show all work and explain your reasoning